

Date: 3/12/2019

Topic: Geometry Streaming Server / Reference Renderer Upgrade

POC: Alex Menzies
menzies@jpl.nasa.gov
626-241-5187

Objective: Support rendering many independent mesh products simultaneously in a manner that allows them to be toggled on and off individual or in groups

Task:

Support for specifying transforms on tilesets (allows mesh products to be tiled in their native reference frame and still be placed correctly with respect to one another)

Upgrade renderer to allow many tilesets to share a common tile cache (reduces risk of crashes due to running out of memory)

Upgrade renderer to prioritize tile requests from multiple tilesets simultaneously (ensures most important tiles are loaded first - prevents one tileset from starving out another)

Add an API to the renderer to support loading collections of tilesets - includes definition of a format that can be used to specify collections / groups of meshes. API will support toggling individual products and groups of products on and off

Generation of test dataset, performance profiling, and optimization

Budget: \$55k

Schedule: TBD